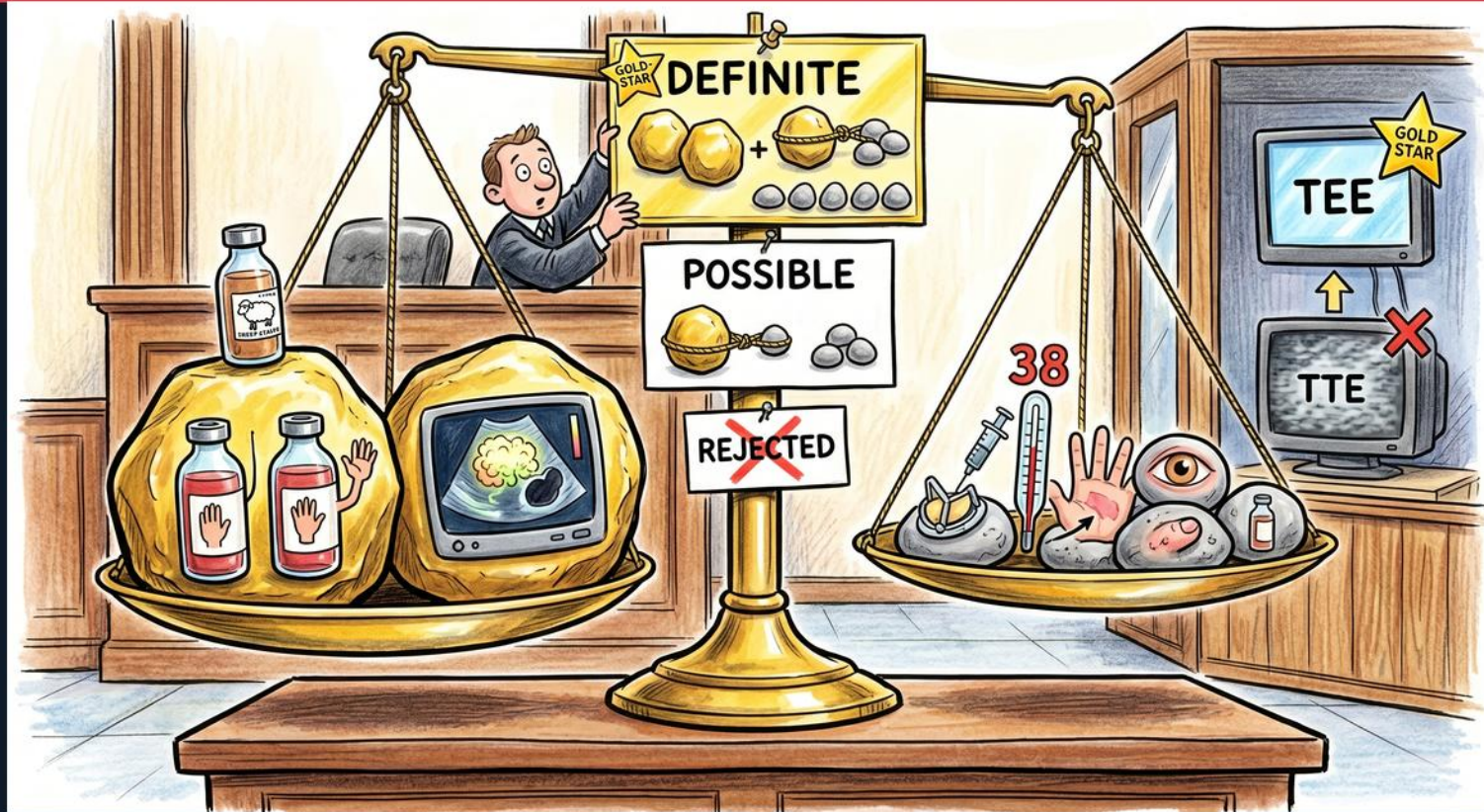


Infective Endocarditis — Duke Criteria





Balance scale = weigh the criteria

WHAT YOU SEE

The giant justice scale weighing heavy major evidence against light minor evidence is the Duke tally that classifies IE.

WHAT IT ENCODES

Duke criteria sort patients into DEFINITE, POSSIBLE, or REJECTED by weighing MAJOR vs MINOR findings. Major = a typical organism in 2 separate cultures (or the *Coxiella* exception) AND endocardial involvement on imaging (vegetation, abscess, fistula, new prosthetic dehiscence, or new regurgitation). The five minor categories are predisposition, fever $\geq 38^{\circ}\text{C}$, vascular phenomena, immunologic phenomena, and microbiologic evidence not meeting major.



DEFINITE plaque (gold star)

WHAT YOU SEE

The starred DEFINITE plaque showing an organism plus stones at the top of the heavy arm is the highest certainty tier.

WHAT IT ENCODES

DEFINITE clinical IE = ANY of: 2 MAJOR, OR 1 major + 3 minor, OR 5 minor criteria. Pathologic definite = organisms or active endocarditis demonstrated on the valve specimen at surgery/autopsy. The combinations are deliberately swappable on exams, so lock them in.



5 minor = definite (man lifting)

WHAT YOU SEE

The man straining to push the DEFINITE side up shows that enough minor criteria alone can still tip into definite IE.

WHAT IT ENCODES

FIVE minor criteria with NO major still equals DEFINITE IE. Don't dismiss a culture-negative, echo-negative patient — count predisposition, fever, vascular, immunologic, and microbiologic minors; five of them clinch the diagnosis without any major criterion.

THE BOARD TRAP

'You must have at least one major criterion to call it definite IE'
-> wrong; five minor criteria with zero major still equals definite IE, so the culture- and echo-negative patient should not be written off.



POSSIBLE plaque

WHAT YOU SEE

The mid-hanging POSSIBLE plaque is the intermediate tier that demands more workup, not dismissal.

WHAT IT ENCODES

POSSIBLE IE = 1 major + 1 minor, OR 3 minor criteria. This intermediate category is suggestive but not definite and **MANDATES** continued evaluation — repeat cultures and TEE — to reclassify as definite or rejected.

THE BOARD TRAP

'Possible IE means stop the workup / no echo needed' -> wrong; possible IE specifically requires further evaluation (TEE, repeat cultures) — 3 minor criteria is possible, not negative.



REJECTED (red-X)

WHAT YOU SEE

The REJECTED plaque crossed out by a red X is the tier where IE is ruled out.

WHAT IT ENCODES

REJECTED IE = a firm alternative diagnosis explains the picture, OR symptoms resolve with ≤ 4 days of antibiotics, OR no pathologic evidence of IE at surgery/autopsy after ≤ 4 days of antibiotics, OR criteria for possible/definite are not met.

THE BOARD TRAP

'Any IE workup that is culture-negative gets rejected' -> wrong; rejection requires a firm alternative diagnosis or resolution within ≤ 4 days of antibiotics, not merely negative cultures — culture-negative IE is real and pursued by serology.



Major: typical organism x2 sets

WHAT YOU SEE

The organism-plus-stones icon on the DEFINITE plaque is a typical IE bug grown from two separate blood-culture sets standing as a major criterion.

WHAT IT ENCODES

MAJOR microbiologic criterion: a TYPICAL IE organism from 2 SEPARATE blood-culture sets — viridans strep, *S. gallolyticus* (bovis), HACEK, *S. aureus*, or community-acquired enterococcus without a primary focus — OR persistently positive cultures (≥ 2 sets >12 h apart, or majority of ≥ 3). Draw 3 sets from separate venipunctures BEFORE antibiotics; IE bacteremia is continuous so fever timing is irrelevant.

THE BOARD TRAP

'One positive culture for a typical organism is a major criterion'
-> wrong; a single positive set is only a MINOR microbiologic criterion — you need 2 separate sets (the lone exception being *Coxiella*).



Coxiella vials — single = major

WHAT YOU SEE

The sheep-labeled serology vials sitting heavy on the major pan are the Coxiella/Q-fever exception that counts as a standalone major criterion.

WHAT IT ENCODES

Coxiella burnetii (Q fever) is the special microbiologic exception: a SINGLE positive blood culture OR a phase I IgG titer $\geq 1:800$ is a STANDALONE major Duke criterion, because Coxiella is culture-negative and diagnosed serologically. Exposure clue: livestock (cattle, sheep, goats).

THE BOARD TRAP

'Coxiella needs 2 positive cultures like other typical organisms'
-> wrong; for Coxiella a single positive culture or phase I IgG $\geq 1:800$ alone is major — with farm-animal exposure, send serology rather than waiting on cultures.



Bartonella serology = major

WHAT YOU SEE

The second serology vial on the major pan is high-titer Bartonella IgG serving as a major criterion.

WHAT IT ENCODES

Bartonella (*B. henselae* from cats, *B. quintana* from homelessness/body lice) is culture-negative and diagnosed serologically; high-titer IgG ($\geq 1:800$) or positive PCR serves as a major microbiologic criterion under the modified/ISCVID Duke criteria.

THE BOARD TRAP

'Negative blood cultures rule out Bartonella IE' -> wrong; Bartonella is culture-negative, so diagnosis relies on serology/PCR — with cat exposure (*henselae*) or homelessness/lice (*quintana*), send serology, not just repeat cultures.



Echo vegetation = major imaging

WHAT YOU SEE

The echo screen showing a vegetation on the heavy major pan is the imaging major criterion.

WHAT IT ENCODES

MAJOR imaging criterion: echo evidence of endocardial involvement — VEGETATION, abscess, pseudoaneurysm, intracardiac fistula, valve perforation, new prosthetic dehiscence, OR new valvular regurgitation. The 2023 Duke-ISCVID update also accepts abnormal periprosthetic PET/CT activity (beyond ~3 months post-op) and paravalvular CT defects as major imaging.

THE BOARD TRAP

'Worsening of a pre-existing murmur counts as the major imaging criterion' -> wrong; only NEW regurgitation on echo or a structural lesion qualifies — an old murmur getting louder does not.



TEE > TTE (gold-star screen)

WHAT YOU SEE

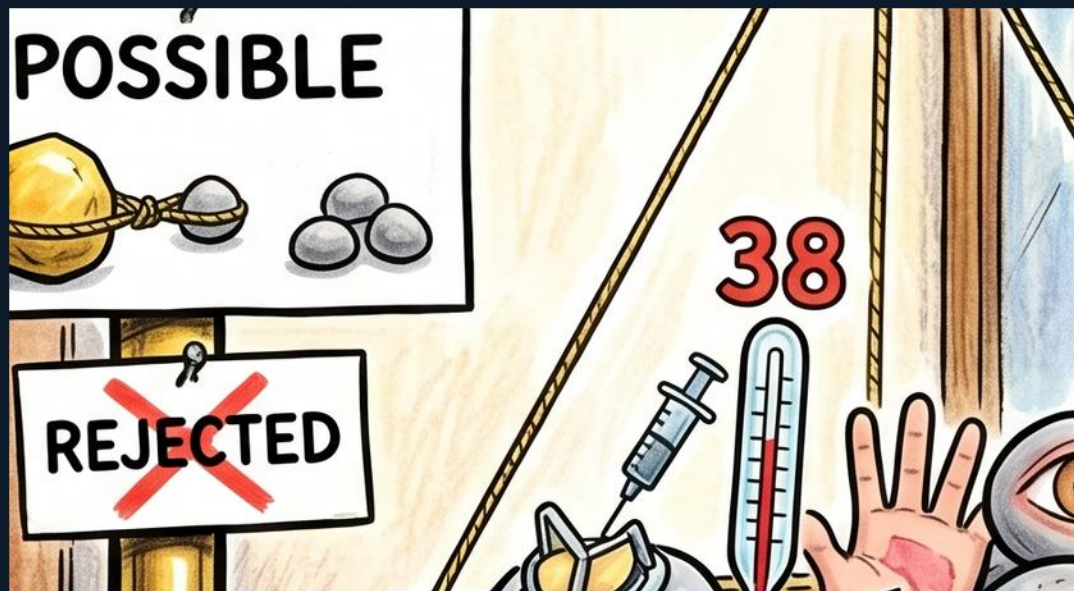
The starred TEE screen with an arrow up from the red-X'd TTE screen is the imaging algorithm escalating from TTE to TEE.

WHAT IT ENCODES

Start with TTE — fast, noninvasive, but sensitivity only ~50–70% for native valves (lower for prosthetics). If TTE is non-diagnostic with ongoing suspicion, or for PROSTHETIC valves, suspected abscess/perivalvular extension, or device leads, go to TEE (>90% sensitivity, the test of choice). Repeat TEE in 3–7 days if initially negative but suspicion persists.

THE BOARD TRAP

'A negative TTE rules out endocarditis' -> wrong; a normal TTE does NOT exclude IE — escalate to TEE and repeat it if still suspicious, and go straight to TEE up front for prosthetic valves or suspected abscess.



Predisposition (minor)

WHAT YOU SEE

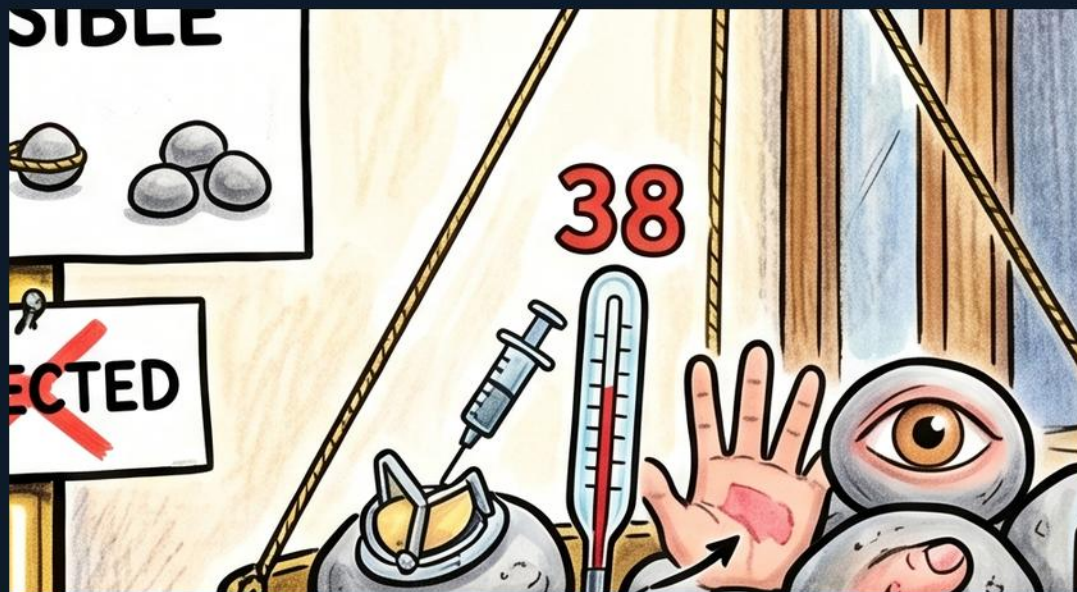
The syringe on the light minor pan represents the predisposition minor criterion — IV drug use or a predisposing heart condition.

WHAT IT ENCODES

MINOR criterion #1 — Predisposition: a predisposing cardiac condition (prosthetic valve, prior IE, congenital heart disease, significant regurgitant/stenotic lesions, HCM) OR injection drug use. This is the structural/behavioral substrate that lets organisms seed the endocardium.

THE BOARD TRAP

'IV drug use is a vascular phenomenon' -> wrong; IVDU and predisposing cardiac conditions are the PREDISPOSITION minor (substrate) — don't double-count it as a vascular phenomenon (which covers embolic/immune sequelae).



Fever $\geq 38^{\circ}\text{C}$ (minor)

WHAT YOU SEE

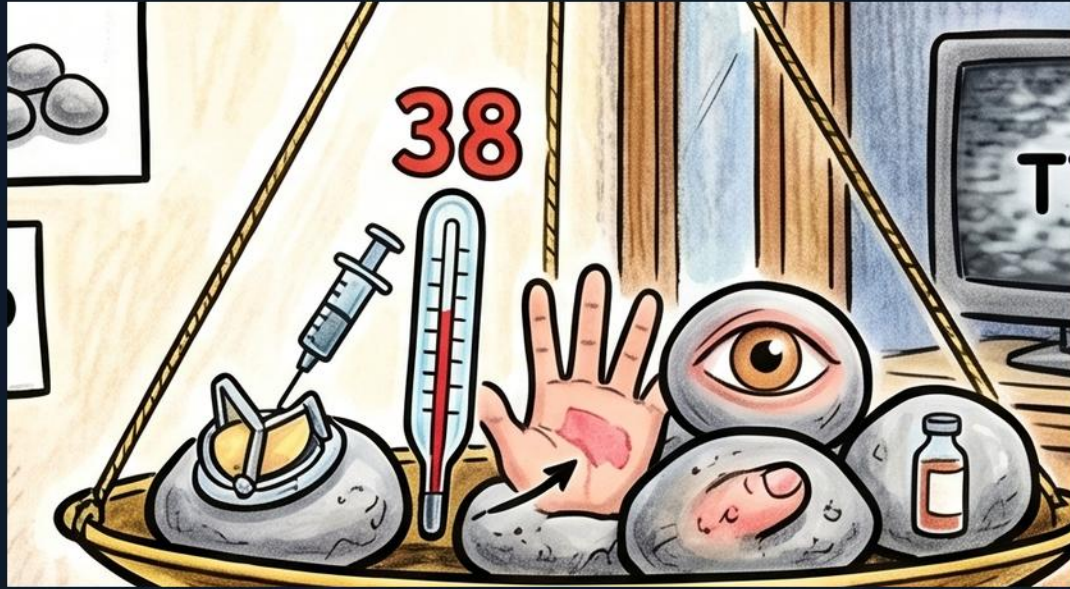
The thermometer reading 38 on the minor pan is the fever minor criterion.

WHAT IT ENCODES

MINOR criterion #2 — Fever: temperature $\geq 38.0^{\circ}\text{C}$ (100.4°F), present in $\sim 90\%$ of IE and the easiest minor to satisfy, but nonspecific on its own. Its absence does not exclude IE.

THE BOARD TRAP

'Absence of fever excludes IE' -> wrong; subacute IE (viridans strep) and elderly/immunosuppressed or recently-antibiotic-treated patients may be afebrile — fever counts only when present, but its absence never rules IE out.



Vascular vs immunologic hands/eye

WHAT YOU SEE

On the minor pan the painless palm (Janeway) sits with the raised tender-pad hand (Osler) and the Roth-spot eye, separating vascular from immunologic minors.

WHAT IT ENCODES

Two distinct minors. **VASCULAR** (embolic): arterial emboli, septic pulmonary infarcts, mycotic aneurysm, intracranial/conjunctival hemorrhage, **JANEWAY** lesions (painless macules on palms/soles). **IMMUNOLOGIC**: glomerulonephritis, **OSLER** nodes (painful finger/toe pulp), **ROTH** spots (retinal hemorrhages, pale centers), rheumatoid factor.

THE BOARD TRAP

'Janeway lesions and Osler nodes are both immunologic' -> wrong; **JANEWAY** = painless, **VASCULAR** (septic emboli), palms/soles, while **OSLER** = painful, **IMMUNOLOGIC**, finger/toe pulp — 'Osler = Ouch (immune); Janeway = painless (vascular).'



Single culture / not-major micro (minor)

WHAT YOU SEE

The lone serology vial on the light minor pan is microbiologic evidence that falls short of a major criterion.

WHAT IT ENCODES

MINOR criterion #5 — Microbiologic evidence not meeting a major: e.g., a single positive blood culture (excluding common skin contaminants and the Coxiella exception), or serologic evidence of active infection with an IE-consistent organism that doesn't reach the major threshold.

THE BOARD TRAP

'A single positive culture for coagulase-negative staph is automatically a criterion' -> wrong; common skin contaminants (CoNS, diphtheroids) in a single set are usually dismissed as contamination — a single positive counts as minor only when the organism is plausibly causative.

Summary — every symbol

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MAJOR imaging criterion: echo evidence of endocardial involvement — VEGETATION, abscess, pseudoaneurysm, intracardiac fistula, valve perforation, new prosthetic dehiscence, OR new valvular regurgitation

11. Predisposition (minor)

MINOR criterion #1 — Predisposition: a predisposing cardiac condition (prosthetic valve, prior IE, congenital heart disease, significant regurgitant/stenotic lesions, HCM) OR injection drug use

13. Vascular vs immunologic hands/eye

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2. DEFINITE plaque (gold star)

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POSSIBLE IE = 1 major + 1 minor, OR 3 minor criteria

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Bartonella (B

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Start with TTE — fast, noninvasive, but sensitivity only ~50–70% for native valves (lower for prosthetics)

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